

U4-D4 Lesson – Rational Exponents

There will be a **quiz next class**, including the basics of this new lesson.

As we've seen before, **rational** refers to fractions (aka, ratios). So, rational exponents are fractional exponents. We're largely concerned with what they mean, which we show by converting between relevant notations.

Rational exponents are just another way to write n^{th} roots! Literally, that's it.

$$x^{\frac{a}{b}} = \sqrt[b]{x^a}$$

Ex.1 Convert to radical notation.

a) $x^{\frac{5}{7}}$

b) $y^{\frac{4}{3}}$

c) $-x^{\frac{2}{5}}$

Ex.2 Convert to exponent notation.

a) $\sqrt[3]{x^{19}}$

b) $\sqrt[16]{x^5}$

c) $\sqrt{x}$

An improper rational exponent corresponds to a radical that can be simplified. There is an interesting connection between the corresponding mixed number and the simplified radical.

Ex.3 $x^{\frac{25}{6}}$

Rational exponents are an opportunity to make you practice fraction operations.

Ex.4 Simplify completely.

a) $\left(x^{\frac{3}{8}}\right)^{\frac{28}{9}}$

b) $x^{\frac{3}{16}} \cdot x^{\frac{5}{6}}$

What's the point of having a new way to write radicals? It means we don't need to invent new ways to deal with radicals; we can use tools we already have in place for exponents.

Ex.5 Simplify completely. Answer in radical form.

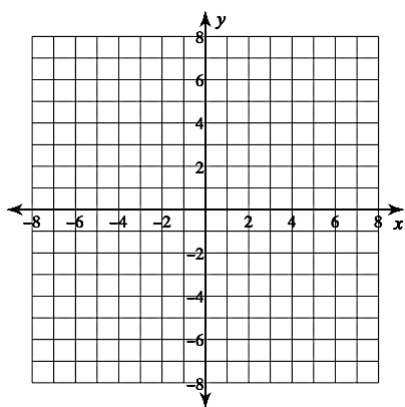
$$\sqrt[8]{x} \cdot \sqrt[12]{x^7}$$

What's on the Quiz?

- Graphing square roots, cube roots and cubic functions
- Simplifying radicals, including n^{th} roots
- Converting between radical and exponent notation (Ex.1 and Ex.2)

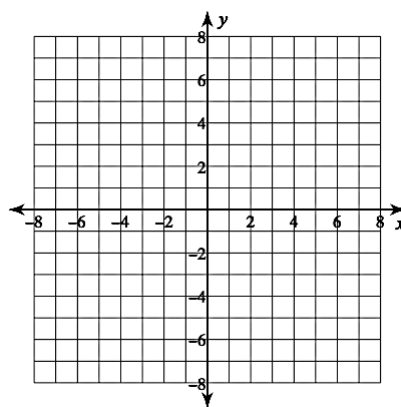
Ex.6 Graph. State starting point, domain and range.

$$f(x) = 2\sqrt{3-x} - 4$$



Ex.7 Graph. State starting point, domain and range.

$$f(x) = -(-x - 4)^3 + 2$$



Ex.8 Simplify completely. Show work on a.

a) $\sqrt[6]{576}$

b) $\sqrt{x^{25}}$

c) $\sqrt{x^{12}y^{15}}$

d) $\sqrt[4]{x^{19}}$

e) $\sqrt[7]{x^{11}y^{20}}$

f) $\frac{1}{\sqrt[9]{x}}$